

Integer Arithmetic



BENNETT
UNIVERSITY
TIMES OF INDIA GROUP

Linear Congruence

Inverses

Chinese Remainder Theorem

An expression of the form $ax \equiv b \pmod{m}$ where $a \not\equiv 0 \pmod{m}$ i.e m does not divide a , is called a linear congruence modulo m .

Solution Let $d = \gcd(a, m)$

- i) The linear congruence has a solution if and only if $d \mid b$ and there is no solution otherwise.
- ii) The solution of the linear congruence can be obtained by solving the following congruence:
 $(a/d)x \equiv (b/d) \pmod{(m/d)}$
- iii) The given congruence has d solutions which are mutually incongruent modulo m .

Note: Let x_0 be the unique smallest positive solution, then $x_0, x_0 + (m/d), x_0 + 2(m/d), \dots, x_0 + (d-1)(m/d)$ will be the d solutions of the given congruence.

Find the solution of the congruence

$$6x \equiv 3 \pmod{9}$$

Solution:

$\gcd(6,9)=3$ and 3 divides 3.

Thus 3 solutions are possible.

By step ii)

$$2x \equiv 1 \pmod{3}$$

Choosing $x=0,1,2$ and testing the congruence, we get $x_0 = 2$

We have $m=9$ and $d=3$

Thus the other two solution are

$$x_1 = x_0 + (m/d) = 2 + 3 = 5$$

$$x_2 = x_0 + 2(m/d) = 2 + 6 = 8$$

Hence 2, 5 and 8 is the solution.

1. Find the solution of the congruence $2x \equiv 1 \pmod{5}$.
2. Find the solution of the congruence $3x \equiv 2 \pmod{4}$
3. Find the solution of the congruence $4x \equiv 3 \pmod{2}$
4. Find the solution of the congruence $234x \equiv 60 \pmod{762}$

Definition of Congruence:

If a and b are integers and m is a positive integer, then a is congruent to b modulo m if m divides $(a-b)$.

$$a \equiv b \pmod{m}$$

$$a \not\equiv b \pmod{m}$$

A) Additive Inverse

In Z_n , two numbers a and b are additive inverses of each other if $a + b \equiv 0 \pmod{n}$

Therefore, in Z_n , additive inverse of a can be calculated as $b = n - a$.

e.g. Additive inverse of 4 in Z_{10} is $10 - 4 = 6$

Find all additive inverse pairs in Z_{10}

Solution: $(0,0), (1,9), (2,8), (3,7), (4,6), (5,5)$

.

B) Multiplicative Inverse

In Z_n , two numbers a and b are multiplicative inverses of each other if $a \times b \equiv 1 \pmod{n}$

e.g. Multiplicative inverse of 3 is 7 in Z_{10}

Note: a has a multiplicative inverse in Z_n , if and only if $\gcd(n,a)=1$.

In this case, a and n are said to be relatively prime.

Try Yourself

1. Find the multiplicative inverse of 8 in Z_{10}
2. Find all the inverses in Z_{10}

The Chinese Remainder Theorem (CRT) is used to solve a set of congruent equations with one variable but different moduli, which are relatively prime as shown below:

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$x \equiv a_3 \pmod{m_3}$$

$$\dots x \equiv a_k \pmod{m_k}$$

CRT states that the above equations have a unique solution if the moduli are relatively prime.

1. Find $M = m_1 \times m_2 \times m_3 \dots \times m_k$. This is the common modulus.
2. Find $M_1 = M/m_1, M_2 = M/m_2, M_3 = M/m_3, \dots, M_k = M/m_k$
3. Find the multiplicative inverses of $M_1, M_2, \dots, M_k$ using the corresponding moduli ($m_1, m_2, \dots, m_k$.)
Call the inverses $M_1^{-1}, M_2^{-1}, \dots, M_k^{-1}$.
4. The solution to the simultaneous equations is
$$x = (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + \dots + a_k M_k M_k^{-1}) \bmod M$$

Find the solution to the simultaneous equations

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

So $a_1 = 2, a_2 = 3, a_3 = 2$

1. $m_1 = 3, m_2 = 5, m_3 = 7$

2. $M_1 = M/m_1 = 105/3 = 35$

$$M_2 = M/m_2 = 105/5 = 21$$

$$M_3 = M/m_3 = 105/7 = 15$$

3. $M_1^{-1} : 35 \times () \equiv 1 \pmod{3}$ The inverses are

$$M_1^{-1} = 2, M_2^{-1} = 1, M_3^{-1} = 1 \text{ (use trial and error)}$$

$$4. x = (a_1 \times M_1 \times M_1^{-1} + a_2 \times M_2 \times M_2^{-1} + a_3 \times M_3 \times M_3^{-1}) \pmod{M}$$

$$= (2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1) \pmod{M}$$

$$= (140 + 63 + 30) \pmod{105}$$

$$= 233 \pmod{105}$$

$$= 23$$

Find an integer that has a remainder of 3 when divided by 7 and 13 but is divisible by 12 using CRT.

Solution

$$x \equiv 3 \pmod{7}$$

$$x \equiv 3 \pmod{13}$$

$$x \equiv 0 \pmod{12}$$

$$x=276$$